BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 068 · Records and Structured Data

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. How is a cell selected in a two-dimensional array?

Q2. What is a router's main job?
A. Store user passwords
B. Print network documents
C. Run spreadsheet formulas
D. Link different networks

Q3. “An IPv4 address is always six pairs of hexadecimal digits separated by colons.” Correct this. Explain why.

Q4. Tablets need to join a wired LAN. “A WAP changes domain names into IP addresses.” Correct this. How does a WAP help the tablets?

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I Do · Read new knowledge

Key vocabulary

Record — Groups related fields about one item or entity, potentially with different data types.

Field — A field is one named property within a record such as name or age.

Data types — A pupil record might store a string name, integer age and real score.

1D array versus record — An array stores indexed values of one type; records group named fields of potentially different types.

Syntax — The rules governing valid programming statements.

Test data — An input used to check expected results.

New learning

Record: Groups related fields about one item or entity, potentially with different data types. Apply this in a pupil profile.

Field: A field is one named property within a record such as name or age. Apply this in a library book.

Data types: A pupil record might store a string name, integer age and real score. Apply this in a sports member.

1D array versus record: An array stores indexed values of one type; records group named fields of potentially different types. Apply this in a patient appointment.

Worked example

A pupil profile is being developed. Explain how record and field are used. [4 marks]

1. Read the scenario and identify the data the program needs.

2. Record [1]; Groups related fields about one item or entity, potentially with different data types. [1].

3. Field [1]; A field is one named property within a record such as name or age. [1].

4. Two linked points for each concept.

We Do · Practise with guidance

Guided 1

A library book uses record. Explain how it helps. [2 marks]

Worked method: Record [1]; Groups related fields about one item or entity, potentially with different data types. [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Guided 2

Explain field in a sports member. [2 marks]

Worked method: Field [1]; A field is one named property within a record such as name or age. [1].

Guided 3

A patient appointment uses data types and 1d array versus record. Explain both. [4 marks]

Worked method: Data types [1]; A pupil record might store a string name, integer age and real score. [1]. 1D array versus record [1]; An array stores indexed values of one type; records group named fields of potentially different types. [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Each row is a pupil record: column 0 is the name; column 1 is the tutor group. Complete the index to print the second pupil's name.

```
array pupils[2,2]
pupils[0,0] = "Ada"
pupils[0,1] = "10A"
pupils[1,0] = "Ben"
pupils[1,1] = "10B"
print(pupils[...])
```

write

Question 1

State TWO relevant techniques from this topic. [2 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

Question 2

Explain how field and data types help with a train passenger. [4 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

Question 3

A programmer designs a school register. Explain why record, data types and 1d array versus record matter. [6 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Exit Ticket · Answer before checking

A record groups related fields about an item, even when the fields have different data types.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Use two indexes: one for the row and one for the column.

Q2. D — Link different networks

Q3. IPv4 supports addressing and routing. It is usually written as four decimal numbers separated by dots.

Q4. A wireless access point connects wireless devices to a network. It has a different job from a switch or router.

I Do / We Do pages · Guided practice

Q1. Record [1]; Groups related fields about one item or entity, potentially with different data types. [1].

Q2. Field [1]; A field is one named property within a record such as name or age. [1].

Q3. Data types [1]; A pupil record might store a string name, integer age and real score. [1]. 1D array versus record [1]; An array stores indexed values of one type; records group named fields of potentially different types. [1].

You Do pages · Code First

```
array pupils[2,2]
pupils[0,0] = "Ada"
pupils[0,1] = "10A"
pupils[1,0] = "Ben"
pupils[1,1] = "10B"
print(pupils[1,0])
```

Expected result: Ben

You Do pages · Independent questions

Q1. Record [1]; Data types [1].

Q2. Field [1]; A field is one named property within a record such as name or age. [1]. Data types [1]; A pupil record might store a string name, integer age and real score. [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Q3. Record [1]; Groups related fields about one item or entity, potentially with different data types. [1]. Data types [1]; A pupil record might store a string name, integer age and real score. [1]. 1D array versus record [1]; An array stores indexed values of one type; records group named fields of potentially different types. [1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: